

Berry and Uhlmann Geometry of Entangled Photon-Added Coherent States under the Global Two-Mode Cycle

$$U(\phi) = e^{-i(\hat{n}_A + \hat{n}_B)\phi}$$

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Abstract

We develop the geometric-phase structure of bipartite photon-added coherent states (PACS) taking as the primary cycle the *free evolution of both modes*, $U(\phi) = e^{-i(\hat{n}_A + \hat{n}_B)\phi}$. This is the evolution an actual propagating two-mode field undergoes, and it is a more natural starting point than a cycle acting on one mode alone: under the global cycle *both* reduced states traverse isospectral orbits and *both* carry Uhlmann holonomy, so the two subsystems are treated on an equal footing from the outset. We obtain in closed form the global Berry phase, which is the total mean photon number and therefore splits exactly into single-mode contributions $\Phi_B = \Phi_B^{(A)} + \Phi_B^{(B)}$; in the balanced symmetric family it is an affine function of the concurrence and exactly twice the single-mode result. For the mixed reduced states we derive the Uhlmann holonomy from the parallel-transport condition in its correct anticommutator form, $[S, \dot{S}] = i\{\rho, \Lambda\}$, which yields the compact statement that parallel transport renormalizes the off-diagonal generator by exactly the concurrence, $(G + \Lambda)_{\text{od}} = C G_{\text{od}}$, and hence $\Phi_U = \text{Arg}[(-1)^\omega \cos \pi b - i \omega (1 - 2p) b^{-1} \sin \pi b]$ with $b = \omega \sqrt{1 + C^2 - 4p(1 - p)}$. The nodal analysis then gives a finite, equally spaced set: each photon-addition separation Δ supports exactly Δ isolated Uhlmann singularities at $C_k = (2k+1)/2\Delta$, $k = 0, \dots, \Delta-1$, all bounded by a critical amplitude α_g defined by $|g_{mn}(\alpha_g)|^2 = 1/2$. We further explain why the two subsystems are *not* Uhlmann-symmetric despite sharing a concurrence: Schmidt isospectrality fixes the length of each reduced Bloch vector but not its orientation relative to that mode's own logical axis, so the local populations p_A, p_B differ whenever $\beta \neq \alpha$; in the crossed family the winding numbers differ in sign instead, producing an exact antisymmetry $\Phi_U^B = -\Phi_U^A$. Every closed form is validated against independent numerics: direct summation in a truncated two-mode Fock space for the Berry sector, and two connection-free holonomy algorithms for the Uhlmann sector. A fully commented Jupyter notebook reproduces all eleven figures and both tables.

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1 Introduction

Photon addition turns a Glauber coherent state into a strongly nonclassical one while preserving an essentially closed analytic structure controlled by a single integer. The photon-added coherent states (PACS) of Agarwal and Tara [1] have since become a standard laboratory for the classical–quantum boundary of light. Entanglement in bipartite PACS was analysed by Domínguez-Serna, Mendieta-Jiménez and Rojas [5], who showed that superpositions of *nonorthogonal* PACS branches reduce, after Gram–Schmidt orthogonalization, to an effective two-qubit problem with a computable Wootters concurrence [4]. In parallel, geometric phases of coherent-state superpositions have been studied for two-mode entangled coherent states [6] and for nonlinear and squeezed states [7], while Uhlmann’s mixed-state holonomy [3] has been developed for coherent states by Wang *et al.* [8].

Why the global cycle. A geometric phase is a property of a *path*, so the first thing to fix is which path. Earlier treatments of this problem cycled a single mode, $U_A(\phi) \otimes I_B$. That choice has a structural drawback: by the cyclicity of the partial trace, a strictly local unitary leaves ρ_B *exactly* invariant,

$$\mathrm{Tr}_A [(U_A \otimes I) \rho_{AB} (U_A^\dagger \otimes I)] = \mathrm{Tr}_A [\rho_{AB}], \quad (1)$$

which is the content of no-signalling. The path of ρ_B is then a *point*, its Uhlmann holonomy is the identity, and the second subsystem drops out of the discussion — not because it lacks geometry, but because the chosen cycle does not probe it.

Here we take instead the *global* cycle

$$U(\phi) = e^{-i(\hat{n}_A + \hat{n}_B)\phi} = U_A(\phi) \otimes U_B(\phi), \quad \hat{n}_s = a_s^\dagger a_s, \quad (2)$$

i.e. free propagation of the two-mode field. It is the physically natural evolution, and under it $\rho_A(\phi) = U_A \rho_A(0) U_A^\dagger$ and $\rho_B(\phi) = U_B \rho_B(0) U_B^\dagger$ both move: both subsystems traverse isospectral loops and both carry nontrivial Uhlmann holonomy. The whole analysis is rebuilt on this basis.

Contributions.

- (i) The closed-form global Berry phase, equal to the total mean photon number, with its exact decomposition into single-mode contributions and its affine dependence on the concurrence (Sec. 4, Figs. 1–5).
- (ii) The Uhlmann connection derived from the parallel-transport condition in anticommutator form, giving the compact result $(G + \Lambda)_{\text{od}} = C G_{\text{od}}$ and the closed phase with $b = \omega\sqrt{1 + C^2 - 4p(1-p)}$ (Sec. 5).
- (iii) A finite nodal set: exactly Δ isolated singularities at $C_k = (2k + 1)/2\Delta$, common to both subsystems, all bounded by α_g (Sec. 6, Figs. 6, 8).
- (iv) An explanation of the $A \leftrightarrow B$ asymmetry: the concurrence is shared but the local populations are not, with the symmetry locus $\beta = \alpha$ and an exact antisymmetry in the crossed family (Sec. 7, Figs. 7, 11).
- (v) A complete independent validation layer and a reproducible notebook (Sec. 8).

2 Photon-added coherent states

A PACS is obtained by applying the creation operator n times to a coherent state and renormalizing, $|\alpha, n\rangle = a^{\dagger n} |\alpha\rangle / [\langle\alpha| a^n a^{\dagger n} |\alpha\rangle]^{1/2}$. Inserting $|\alpha\rangle = e^{-|\alpha|^2/2} \sum_j \alpha^j / \sqrt{j!} |j\rangle$ and using $a^{\dagger n} |j\rangle = \sqrt{(j+n)!/j!} |j+n\rangle$ gives $a^{\dagger n} |\alpha\rangle = e^{-|\alpha|^2/2} \sum_{k \geq 0} \alpha^k \sqrt{(n+k)!/k!} |n+k\rangle$, whose norm is fixed by the PACS identity $\langle\alpha| a^n a^{\dagger n} |\alpha\rangle = n! L_n(-|\alpha|^2)$. Hence

$$|\alpha, n\rangle = \frac{e^{-|\alpha|^2/2}}{\sqrt{n! L_n(-|\alpha|^2)}} \sum_{k=0}^{\infty} \frac{\alpha^k \sqrt{(n+k)!}}{k!} |n+k\rangle, \quad (3)$$

with L_n the ordinary Laguerre polynomial. The mean photon number follows from the Laguerre recursion,

$$\bar{n}_{\alpha,n} = (n+1) \frac{L_{n+1}(-|\alpha|^2)}{L_n(-|\alpha|^2)} - 1, \quad (4)$$

which correctly reduces to $|\alpha|^2$ at $n = 0$.

Under the global cycle (2) each PACS transforms as

$$U(\phi) |\alpha, n\rangle = e^{-in\phi} \left| \alpha e^{-i\phi}, n \right\rangle, \quad (5)$$

because $e^{-i\hat{n}\phi}$ is diagonal in the Fock basis: the coherent amplitude rotates and the photon-addition index contributes an *explicit* phase $e^{-in\phi}$. These index phases are what generate the winding numbers below.

3 Bipartite model and logical-qubit reduction

Since PACS branches are not orthogonal, a bipartite superposition must be orthogonalized before it can be read as a two-qubit state. With p_1, p_2 the local branch overlaps and $N_j = \sqrt{1 - |p_j|^2}$, the Gram–Schmidt bases are

$$|0_s\rangle = (\text{first branch}), \quad |1_s\rangle = \frac{(\text{second branch}) - p_s |0_s\rangle}{N_s}, \quad s = A, B, \quad (6)$$

and the reduced states become rank-2 with a Wootters concurrence $C = 2\sqrt{\det \rho_s}$ [4]. We use two families:

Branch-correlated.

$$|\Phi\rangle \propto \mu |\alpha, m\rangle_A |\beta, m\rangle_B + \nu e^{i\chi} |\alpha, n\rangle_A |\beta, n\rangle_B, \quad (7)$$

whose branch phases under (5) are $e^{-i2m\phi}$ and $e^{-i2n\phi}$; each mode separately winds with $\omega_A = \omega_B = +\Delta$, $\Delta \equiv n - m$.

Crossed (swap).

$$|\Phi\rangle \propto \mu |\alpha, m\rangle_A |\beta, n\rangle_B + \nu e^{i\chi} |\beta, n\rangle_A |\alpha, m\rangle_B. \quad (8)$$

Here the photon-addition indices are *interchanged* between modes, so $\omega_A = +\Delta$ while $\omega_B = -\Delta$: the two subsystems wind in opposite senses.

With $\nu = \sqrt{1 - \mu^2}$, $g_\alpha \equiv g_{mn}(\alpha) = \langle \alpha, m | \alpha, n \rangle$ and $h_\beta \equiv g_{mn}(\beta)$, the branch-correlated family gives

$$\mathcal{N}^2 = 1 + 2\mu\nu g_\alpha h_\beta, \quad p_A = \frac{\nu^2(1 - g_\alpha^2)}{\mathcal{N}^2}, \quad p_B = \frac{\nu^2(1 - h_\beta^2)}{\mathcal{N}^2}, \quad (9)$$

$$\det \rho_A = \det \rho_B = \frac{\mu^2 \nu^2 (1 - g_\alpha^2)(1 - h_\beta^2)}{\mathcal{N}^4} = \left(\frac{C}{2}\right)^2, \quad (10)$$

while the crossed family, with $s \equiv \langle \alpha, m | \beta, n \rangle$, gives $\mathcal{N}^2 = 1 + 2\mu\nu s^2$, $p_A = p_B = \nu^2(1 - s^2)/\mathcal{N}^2$ and $C = 2\mu\nu(1 - s^2)/\mathcal{N}^2$; there $p_A = p_B$ holds identically because the two branches are exchanged copies of one another, so $N_1 = N_2$.

Equation (10) is worth emphasizing: for a bipartite *pure* state the Schmidt decomposition makes ρ_A and ρ_B isospectral, so the determinant — and hence the concurrence — is *shared* by the two subsystems. This will be central in Sec. 7.

4 Global Berry phase

The Berry connection of the global cycle is the total mean photon number,

$$\mathcal{A}_\phi = \langle \Phi | (\hat{n}_A + \hat{n}_B) | \Phi \rangle, \quad (11)$$

independent of ϕ since the generator commutes with itself, so $\Phi_B = \int_0^{2\pi} \mathcal{A}_\phi d\phi = 2\pi \langle \hat{n}_A + \hat{n}_B \rangle$. Expanding (7) and using $\kappa_\alpha \equiv \langle \alpha, m | a^\dagger a | \alpha, n \rangle$ (Appendix B),

$$\frac{\Phi_B}{2\pi} = \frac{\mu^2(\bar{n}_{\alpha,m} + \bar{n}_{\beta,m}) + \nu^2(\bar{n}_{\alpha,n} + \bar{n}_{\beta,n}) + 2\mu\nu \cos \chi (\kappa_\alpha h_\beta + g_\alpha \kappa_\beta)}{1 + 2\mu\nu \cos \chi g_\alpha h_\beta} \quad (12)$$

Exact mode decomposition. Because the generator is a *sum* of commuting single-mode operators, the phase splits exactly,

$$\Phi_B = \Phi_B^{(A)} + \Phi_B^{(B)}, \quad \frac{\Phi_B^{(A)}}{2\pi} = \langle \hat{n}_A \rangle, \quad \frac{\Phi_B^{(B)}}{2\pi} = \langle \hat{n}_B \rangle, \quad (13)$$

with $\langle \hat{n}_A \rangle = [\mu^2 \bar{n}_{\alpha,m} + \nu^2 \bar{n}_{\alpha,n} + 2\mu\nu \cos \chi \kappa_\alpha h_\beta]/\mathcal{N}^2$ and the mirror expression for B . This decomposition, verified to 10^{-16} in Sec. 8, is displayed in Fig. 5: for $\beta \neq \alpha$ the two modes contribute unequally, and the mode whose amplitude is held fixed contributes a nearly constant amount.

Symmetric family and concurrence dependence. For $\beta = \alpha$ the two contributions coincide and Φ_B is exactly *twice* the single-mode result. In the balanced symmetric case ($\mu = \nu$, $\chi = 0$), writing $g^2 = (1 - C)/(1 + C)$,

$$\frac{\Phi_B}{2\pi} = (1 + C) \bar{n}_{\text{av}} + (1 - C) Q_{mn}, \quad \bar{n}_{\text{av}} = \frac{1}{2}(\bar{n}_{\alpha,m} + \bar{n}_{\alpha,n}), \quad (14)$$

again *affine in the concurrence*, interpolating between the mean-photon average at $C = 0$ and the interference term Q_{mn} at $C = 1$, and again exactly twice the one-mode expression.

Branch-phase interference. The cycle in ϕ produces a connection that already contains the coherent cross term κg , yet no oscillation in ϕ : the interference is real but hidden inside a constant. Introducing the tunable relative branch phase χ of Eqs. (7) exposes it as an explicit 2π -periodic modulation, Fig. 3.

5 Uhlmann holonomy of the reduced states

Under the global cycle both reduced states follow isospectral orbits

$$\rho_s(\phi) = V_s(\phi) \rho_s(0) V_s^\dagger(\phi), \quad V_s(\phi) = e^{i \frac{\omega_s}{2} \sigma_z \phi}, \quad s = A, B. \quad (15)$$

5.1 The parallel-transport condition

With $\rho = W W^\dagger$, $W = \sqrt{\rho} U$, $S \equiv \sqrt{\rho}$ and $\dot{U} = -i \Lambda U$, one has $W^\dagger \dot{W} = U^\dagger (S \dot{S} - i \rho \Lambda) U$. Uhlmann's condition is that $W^\dagger \dot{W}$ be Hermitian, which gives

$$\boxed{[S, \dot{S}] = i \{\rho, \Lambda\}} \quad (16)$$

an *anticommutator* (Sylvester) equation. In the eigenbasis of ρ , with eigenvalues λ_j , it is solved elementwise by

$$\Lambda_{jk} = \frac{-i [S, \dot{S}]_{jk}}{\lambda_j + \lambda_k}. \quad (17)$$

We stress that the superficially similar sandwich expression $\Lambda = \frac{1}{2i} S^{-1} [S, \dot{S}] S^{-1}$, whose elements carry a denominator $2\sqrt{\lambda_j \lambda_k}$, does *not* solve Eq. (16) unless $\lambda_j = \lambda_k$; the two coincide only for the maximally mixed state. Section 8 confirms Eq. (17) against two connection-free numerical holonomies.

5.2 Closed form

For the isospectral loop, $\dot{S} = i[G, S]$ with $G = \frac{\omega}{2} \sigma_z$, so $[S, \dot{S}]_{jk} = -i G_{jk} (\sqrt{\lambda_j} - \sqrt{\lambda_k})^2$ and Eq. (17) gives a purely off-diagonal connection

$$\Lambda_{jk} = -G_{jk} \frac{(\sqrt{\lambda_j} - \sqrt{\lambda_k})^2}{\lambda_j + \lambda_k}, \quad j \neq k. \quad (18)$$

Since $\text{Tr} \rho = 1$ we have $\lambda_j + \lambda_k = 1$ for $j \neq k$, and $2\sqrt{\lambda_+ \lambda_-} = 2\sqrt{\det \rho} = C$, so $(\sqrt{\lambda_+} - \sqrt{\lambda_-})^2 = 1 - C$ and the connection collapses to

$$\boxed{\Lambda_{\text{od}} = -(1 - C) G_{\text{od}} \quad \Longrightarrow \quad (G + \Lambda)_{\text{od}} = C G_{\text{od}}} \quad (19)$$

the concurrence is exactly the factor by which parallel transport renormalizes the off-diagonal part of the generator. Writing the Bloch vector of $\rho_s(0)$ with $\cos \theta_s = (1 - 2p_s)/r$, $r = \sqrt{1 - C^2}$, and using $e^{iG2\pi} = (-1)^\omega \mathbb{K}$ for integer ω , the holonomy $U_{\text{hol}} = e^{iG2\pi} e^{-i(G+\Lambda)2\pi}$ gives

$$\Phi_U = \text{Arg} \left[(-1)^\omega \cos(\pi b) - i \frac{\omega(1 - 2p)}{b} \sin(\pi b) \right], \quad b = \omega \sqrt{1 + C^2 - 4p(1 - p)} \quad (20)$$

where the compact b follows from $b^2 = \omega^2(\cos^2 \theta + C^2 \sin^2 \theta)$ on substituting $r^2 = 1 - C^2$. Two limits check the result: if G commutes with ρ the state does not move and $\Phi_U = 0$; and the maximally mixed loop is again trivial. At the population-symmetric point,

$$p = \frac{1}{2} \quad \implies \quad b = \omega C, \quad (21)$$

so b is there *proportional* to the concurrence.

6 Nodal singularities

Define the complex Uhlmann amplitude $Z_U = (-1)^\omega \cos(\pi b) - i \omega(1 - 2p)b^{-1} \sin(\pi b)$, so $\Phi_U = \text{Arg } Z_U$. A genuine phase singularity needs both parts to vanish; since $\cos(\pi b) = 0 \Rightarrow \sin(\pi b) \neq 0$, the factor $(1 - 2p)$ must vanish independently. Hence

$$p = \frac{1}{2}, \quad |b| = k + \frac{1}{2}, \quad k \in \mathbb{Z}_{\geq 0} \quad (22)$$

At $p = 1/2$, Eq. (21) gives $|b| = \Delta C$, so the concurrence is quantized to

$$C_k = \frac{2k + 1}{2\Delta}, \quad k = 0, 1, \dots, \Delta - 1 \quad (23)$$

the upper bound enforcing $C_k \leq 1$. Thus **each photon-addition separation Δ supports exactly Δ Uhlmann singularities**, equally spaced in concurrence by $1/\Delta$, isolated, with no accumulation point. Because only $|\omega_s|$ enters and the concurrence is shared, the set (23) is *the same for both subsystems*: which entanglement values are geometrically singular is a property of the pair.

6.1 Nodal coordinates and the critical amplitude

For the symmetric family, $p = 1/2$ in Eq. (9) solves exactly: $\mu = \nu(1 - 2q)$ with $q = g_{mn}(\alpha)^2$, i.e.

$$\mu(q) = \frac{1 - 2q}{\sqrt{1 + (1 - 2q)^2}}, \quad \text{along which} \quad C = 1 - 2q. \quad (24)$$

These relations are purely kinematic — properties of the state, not of the connection. Combining with Eq. (23),

$$q_k = \frac{1 - C_k}{2}, \quad \mu_k = \frac{C_k}{\sqrt{1 + C_k^2}}, \quad |g_{mn}(\alpha_k)|^2 = q_k, \quad (25)$$

with α_k obtained by root-finding on the monotonic function $|g_{mn}(\alpha)|^2$. Defining the critical amplitude by

$$|g_{mn}(\alpha_g)|^2 = \frac{1}{2}, \quad (26)$$

the $p = 1/2$ curve exists only for $\alpha \leq \alpha_g$. Since $C_k \geq 1/(2\Delta) > 0$ we have $q_k < 1/2$ strictly, so *every* node satisfies $\alpha_k < \alpha_g$: the critical amplitude is a genuine **bound** on the nodal region, but — the set being finite — not an accumulation point. Table 3 lists the complete nodal set for representative pairs, each verified to satisfy $p = 1/2$, $|b| = k + 1/2$ and $|Z_U| \lesssim 10^{-14}$ simultaneously.

7 Why the two subsystems are not Uhlmann-symmetric

Since the global state is pure, ρ_A and ρ_B are isospectral and share a concurrence. It is then natural to ask why $\Phi_U^A \neq \Phi_U^B$. The answer is that *the Uhlmann phase is not a function of the entanglement alone*. It depends on three inputs,

$$\Phi_U^s = \Phi_U(C, p_s, \omega_s), \quad (27)$$

of which only the first is symmetric:

ingredient	symmetric under $A \leftrightarrow B$?	reason
$C, \det \rho, r$	yes	Schmidt: identical spectra
p_s	no	not an eigenvalue, but a matrix element in the <i>local</i> basis
ω_s	no (crossed family)	indices interchanged between modes

The decisive one is p_s . Isospectrality fixes the *eigenvalues* of ρ_s , but $p_s = (\rho_s)_{22}$ is not an eigenvalue: it is a matrix element in the Gram–Schmidt basis (6) built from *that mode's own* branches, whose overlap is g_α for A and h_β for B . Nothing forces $g_\alpha = h_\beta$.

Geometric reading. With $\cos \theta_s = (1 - 2p_s)/r$ one has the exact identity

$$b_s = \omega_s \sqrt{\cos^2 \theta_s + C^2 \sin^2 \theta_s}, \quad (28)$$

where θ_s is the *tilt* between the reduced Bloch vector of mode s and the σ_z axis of its own logical basis. Therefore:

Schmidt's theorem fixes the **length** of each reduced Bloch vector (r , common to both sides) but **not its orientation** relative to each mode's local axis. Uhlmann holonomy depends on that orientation.

Entanglement is symmetric; *local nonorthogonality is not*. Concretely, for $(m, n) = (0, 2)$, $\alpha = 1.1$, $\beta = 0.8$, $\mu = 0.6$ the two sides share $C = 0.748027$ exactly while $p_A = 0.4725$, $p_B = 0.5263$, i.e. tilts $\theta_A = 85.2^\circ$ and $\theta_B = 94.6^\circ$. Symmetry is restored exactly when the two local geometries coincide, $g_\alpha = h_\beta$, i.e. $\beta = \alpha$ for real amplitudes; Fig. 11(c) shows that the symmetry locus in the (α, β) plane is precisely the diagonal.

Antisymmetry in the crossed family. There $p_A = p_B$ identically but $\omega_B = -\omega_A$, hence $b_B = -b_A$. Inspecting Eq. (20): $(-1)^{\omega_B} = (-1)^{\omega_A}$ and $\cos(\pi b_B) = \cos(\pi b_A)$, so the real parts agree; while $\omega_B/b_B = \omega_A/b_A$ but $\sin(\pi b_B) = -\sin(\pi b_A)$, so the imaginary parts are opposite. Therefore $Z_U^B = \overline{Z_U^A}$ and

$$\boxed{\Phi_U^B = -\Phi_U^A, \quad \Phi_U^A + \Phi_U^B = 0} \quad (29)$$

exactly, for every α, β, μ, m, n — verified to floating-point zero in Sec. 8 and shown in Fig. 7(c). The entangled pair then accumulates no *net* Uhlmann holonomy even though each half accumulates a large one. Since complex conjugation does not move zeros, the two subsystems also share their nodal set exactly in that family.

8 Numerical validation

Every closed form is checked against an independent construction.

Berry sector: truncated two-mode Fock space. Because the global cycle is diagonal in the Fock basis, $e^{-i(\hat{n}_A + \hat{n}_B)\phi} |j\rangle_A |k\rangle_B = e^{-i(j+k)\phi} |j\rangle_A |k\rangle_B$, the evolution can be applied exactly. We build the full coefficient matrix C_{jk} with 90 levels per mode and evaluate overlaps, matrix elements and $\langle \hat{n}_A + \hat{n}_B \rangle$ by direct summation, with no recourse to hypergeometric identities.

Uhlmann sector: two connection-free holonomies. Method 1 is the discrete polar-decomposition algorithm, which implements the parallel-transport condition $W_j^\dagger W_{j+1} \geq 0$ directly: with $\sqrt{\rho_j} \sqrt{\rho_{j+1}} = V_j P_j$ one has $U_{j+1} = V_j^\dagger U_j$ and $\Phi_U = \text{Arg Tr}[\rho_0 U_{\text{hol}}]$. Method 2 integrates $\dot{U} = -i\Lambda U$ after solving Eq. (16) as a linear system at each step. Neither uses a closed-form connection.

Table 1: Independent validation. “Fock” denotes direct summation in the truncated two-mode space; “polar” and “ODE” the two connection-free holonomy algorithms.

Quantity	Checked against	Max. discrepancy
$\langle \alpha, m \beta, n \rangle$, Eq. (32)	Fock	2×10^{-16}
$\kappa_\alpha = g_{mn} Q_{mn}$, Eq. (34)	Fock	6×10^{-15}
Φ_B , Eq. (12)	Fock	2×10^{-14}
Mode split $\Phi_B = \Phi_B^{(A)} + \Phi_B^{(B)}$	Closed forms	9×10^{-16}
Symmetric case $\Phi_B = 2\Phi_B^{(A)}$	Closed forms	0 (exact)
Affine law, Eq. (14)	Direct evaluation	10^{-10}
p_s, C, ω_s , Eqs. (9)–(10)	Fock + winding fit	10^{-9}
Φ_U , Eq. (20)	Polar holonomy	2×10^{-6}
Φ_U , Eq. (20)	Transport ODE	2×10^{-6}
Tilt identity, Eq. (28)	Eq. (20)	2×10^{-16}
Nodes C_k , Eq. (23)	$ Z_U $ at the node	10^{-14}
Antisymmetry, Eq. (29)	Closed form	0 (exact)

As a sharp discriminant, the nodes predicted by Eq. (23) give $|\text{Tr}[\rho_0 U_{\text{hol}}]| \sim 10^{-15}$ in the ODE holonomy, i.e. genuine zeros, whereas the values $C = 2\Delta/(2k+1)$ that follow from the sandwich connection give $|\text{Tr}[\rho_0 U_{\text{hol}}]| = 0.31, 0.90, 0.94$ for $\Delta = 2, k = 2, 3, 4$ — not zeros at all.

9 Results

All figures use real amplitudes; “balanced” means $\mu = \nu = 1/\sqrt{2}$ and “symmetric” means $\beta = \alpha$.

10 Discussion

Choosing the global cycle organizes the problem into three clean levels.

Level 1 — the connection is a photon number. Because $\hat{n}_A + \hat{n}_B$ generates the cycle and commutes with itself, the Berry connection is constant and equals the total mean photon number. Everything in the Berry sector is therefore smooth, single-valued and additive across modes,

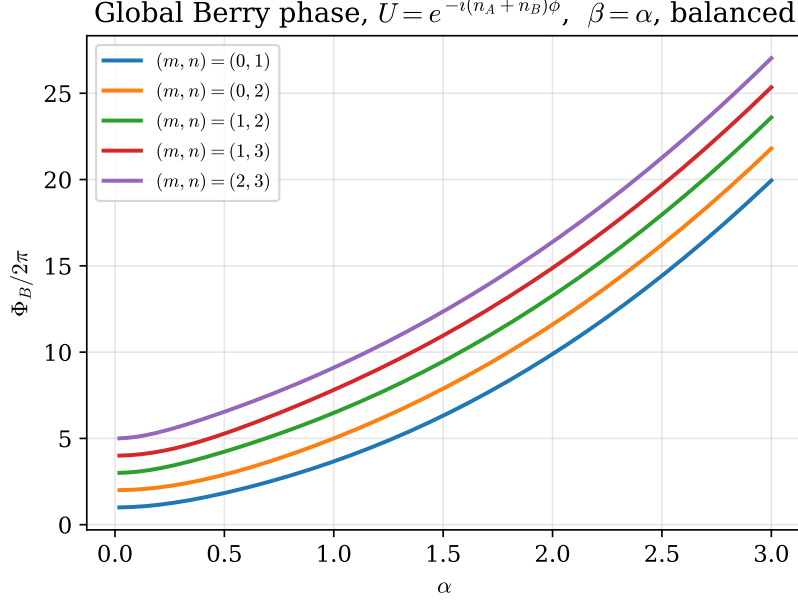


Figure 1: Global Berry phase $\Phi_B/2\pi = \langle \hat{n}_A + \hat{n}_B \rangle$, Eq. (12), versus α for several photon-addition pairs in the balanced symmetric state. Growth is smooth because the connection is constant along the cycle; changing (m, n) shifts the curve through both the diagonal mean-photon terms and the interference term of Eq. (14).

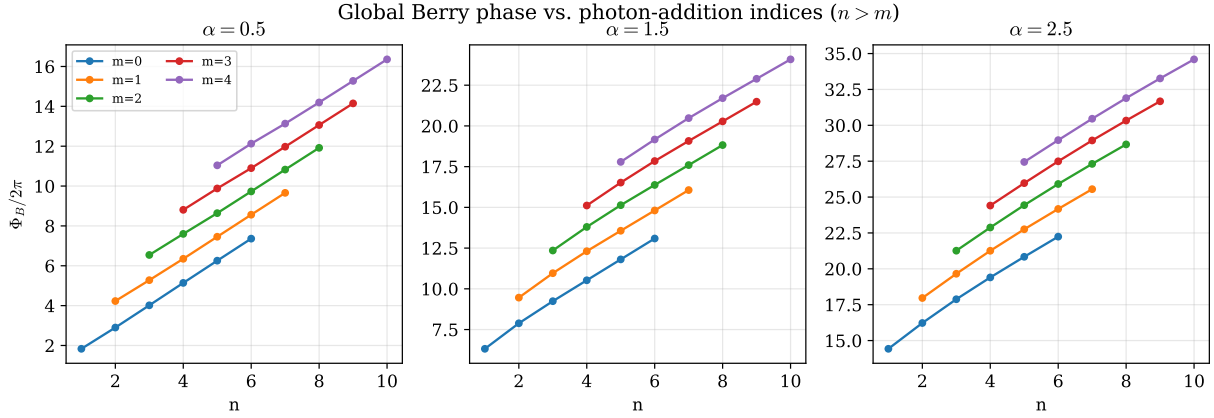


Figure 2: Discrete dependence of the global Berry phase on the photon-addition indices at three coherent amplitudes. Only $n > m$ is shown, the ordering in which the overlap formula (32) is written.

Table 2: Critical amplitudes $\alpha_g(m, n)$ from $|g_{mn}(\alpha_g)|^2 = 1/2$.

$n \backslash m$	0	1	2	3	4
1	1.0000				
2	2.1094	0.7221			
3	3.2756	1.8000	0.5921		
4	4.4576	3.0354	1.5829	0.5135	
5	5.6464	4.2685	2.8269	1.4235	0.4596

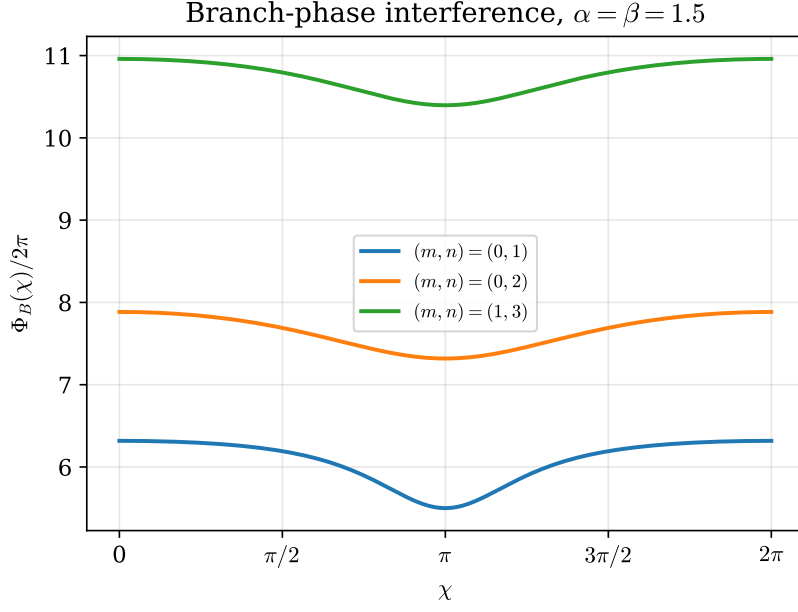


Figure 3: Interference fringes produced by the tunable relative branch phase χ . The ϕ -cycle itself shows no oscillation because its connection, though containing the coherent cross term, is ϕ -independent; χ makes that interference explicit.

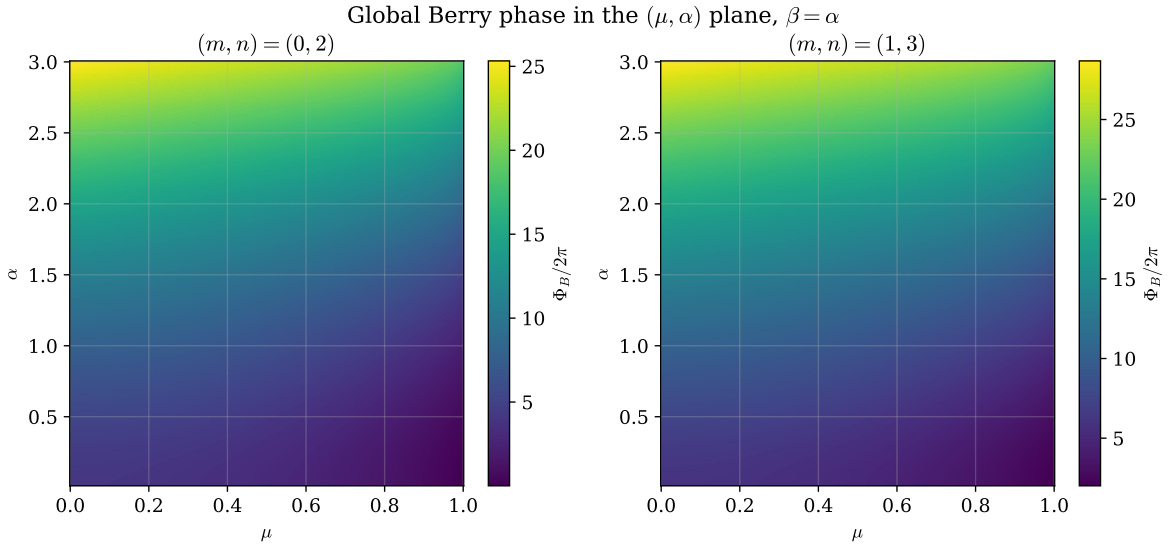


Figure 4: Global Berry phase in the (μ, α) plane. The map is completely smooth: a pure-state Berry phase never develops branch cuts, in contrast with the Uhlmann maps below.

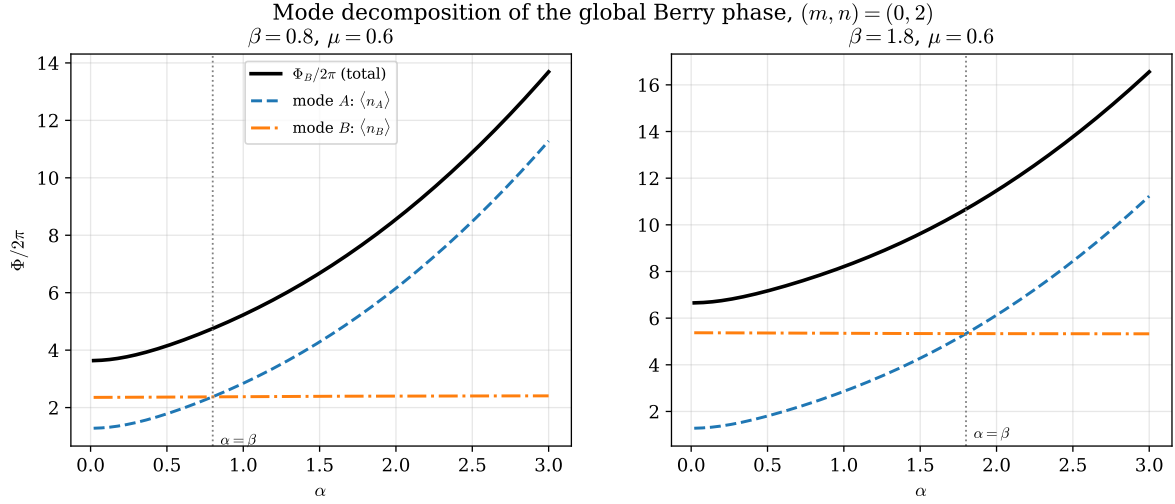


Figure 5: Exact mode decomposition (13) of the global Berry phase for two fixed values of β . The total (black) is the sum of the two single-mode contributions; the mode whose amplitude is held fixed contributes an almost constant amount, so the split is markedly unequal away from $\alpha = \beta$.

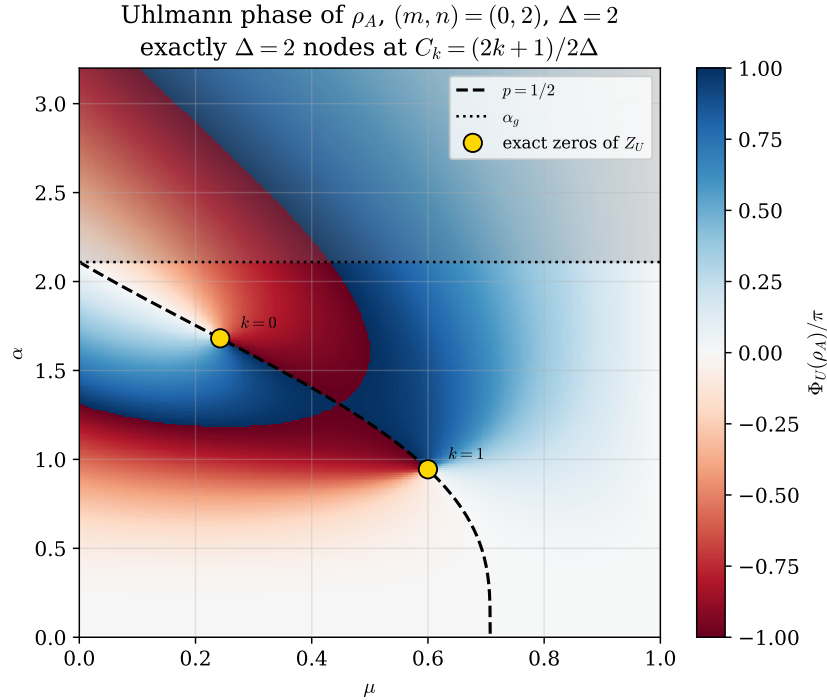


Figure 6: Principal Uhlmann phase of ρ_A in the (μ, α) plane, Eq. (20), for the symmetric family. The dashed curve is the analytic condition $p = 1/2$, Eq. (24); the dotted line is α_g and the shaded region above it cannot contain nodes. Gold circles are the exact zeros of Z_U : for $\Delta = 2$ there are exactly two, labelled $k = 0, 1$, at $C_k = (2k + 1)/2\Delta$.

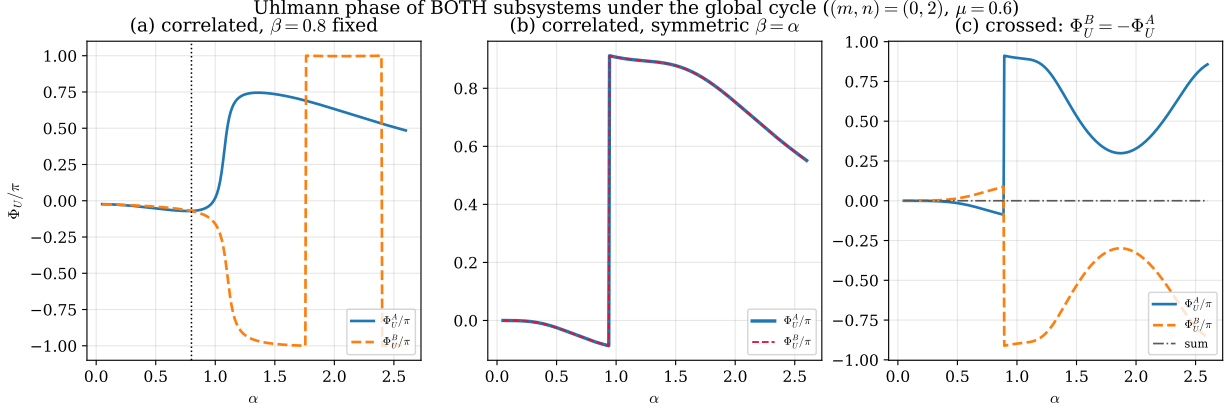


Figure 7: Uhlmann phase of *both* subsystems under the global cycle. (a) Branch-correlated family with $\beta = 0.8$ fixed: the two phases are different curves, meeting only at $\alpha = \beta$, even though they share a concurrence. (b) Symmetric family $\beta = \alpha$: $p_A = p_B$ by Eq. (9) and the curves coincide exactly. (c) Crossed family, where $\omega_B = -\omega_A$: the exact antisymmetry (29) holds and the sum (dash-dotted) vanishes identically.

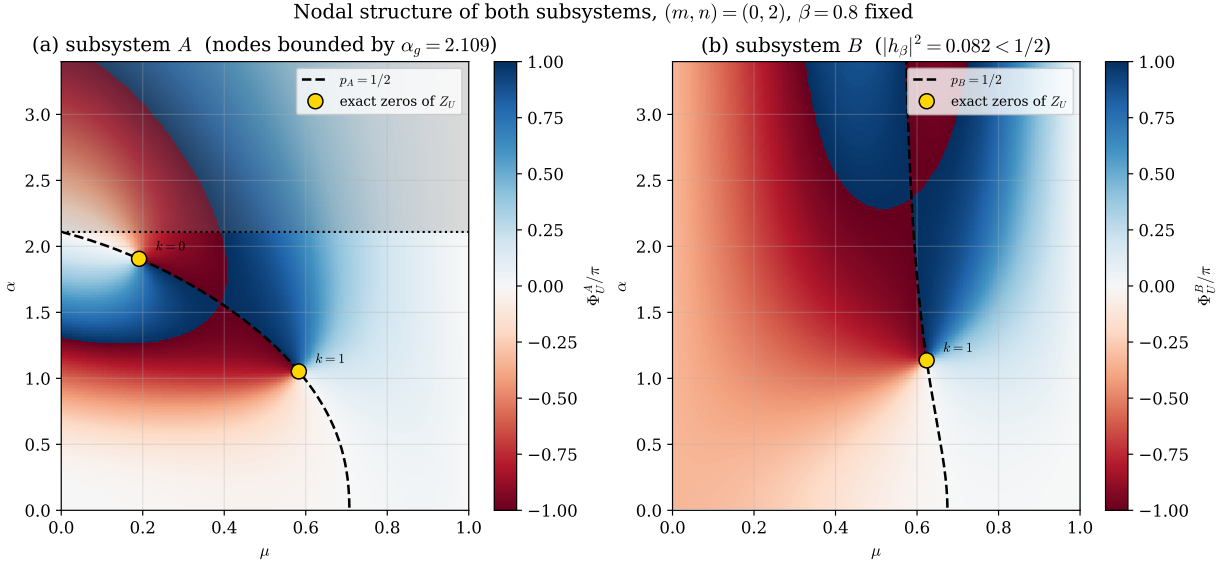


Figure 8: Nodal structure of both subsystems in the (μ, α) plane at fixed $\beta = 0.8$. Both share the same nodal concurrences C_k , Eq. (23), but not their coordinates: the $p_A = 1/2$ curve bends and keeps the A nodes below α_g , whereas the existence criterion for B depends on β alone, so its $p_B = 1/2$ curve is nearly vertical and its nodes are not bounded by α_g (the $k = 0$ node of B lies at $\alpha \approx 5.4$, outside the plotted range).

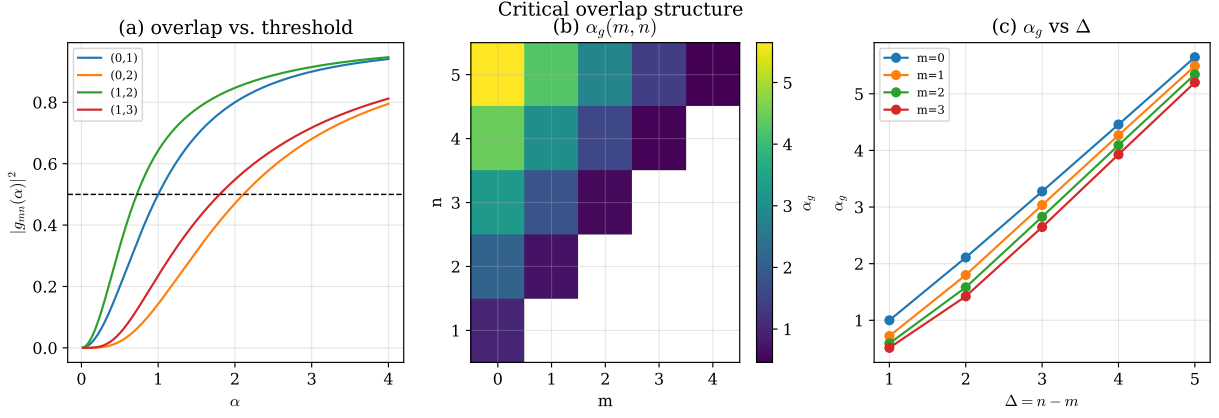


Figure 9: Critical overlap structure. (a) $|g_{mn}(\alpha)|^2$ against the threshold $1/2$. (b) The discrete critical function $\alpha_g(m, n)$ over the grid of Table 2. (c) α_g versus Δ for several fixed m : larger photon-addition separation extends the nodal region to larger coherent amplitude.

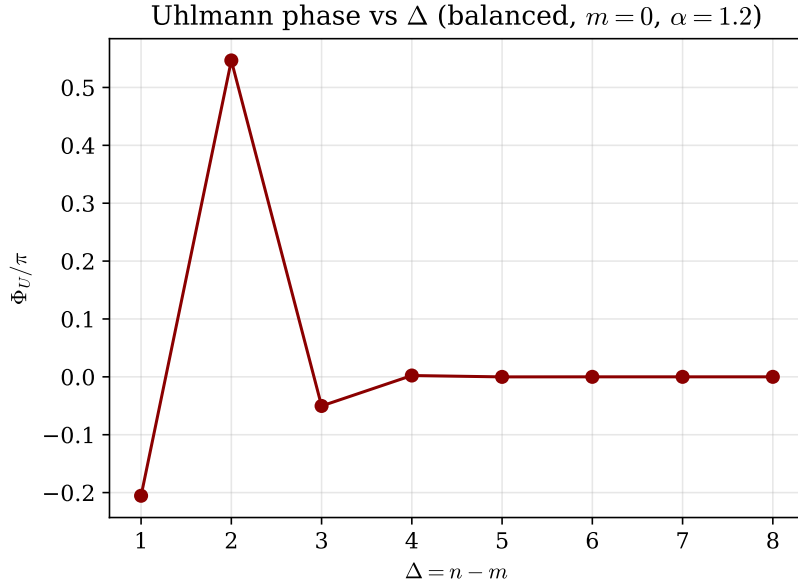


Figure 10: Principal Uhlmann phase versus photon-addition separation Δ for the balanced symmetric state. The nonmonotonic behaviour reflects the dependence of $b = \Delta C$ on Δ through the overlap, combined with principal-branch wrapping of $\text{Arg}(\cdot) \in (-\pi, \pi]$.

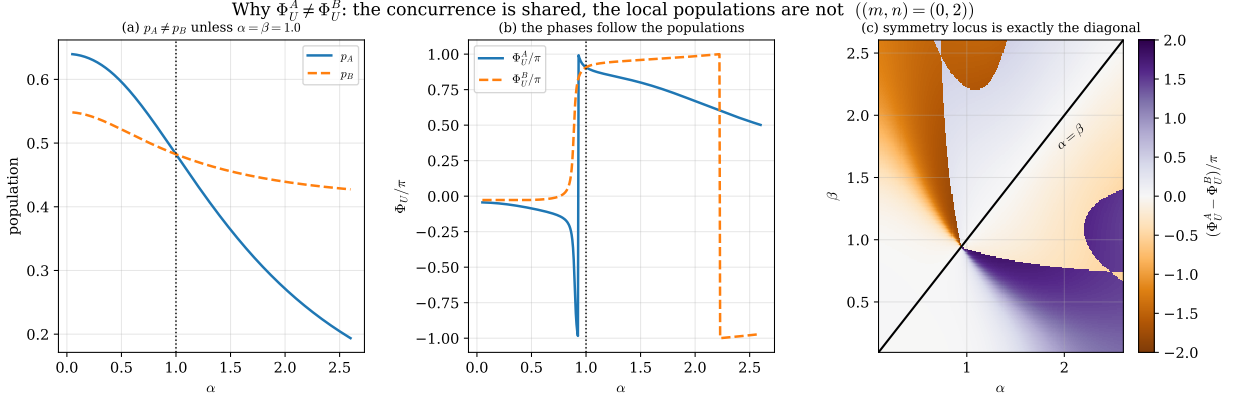


Figure 11: Origin of the $A \leftrightarrow B$ asymmetry (Sec. 7). (a) The local populations differ, crossing only at $\alpha = \beta$. (b) The two Uhlmann phases track those populations. (c) Map of $\Phi_U^A - \Phi_U^B$ in the (α, β) plane: the symmetry locus is exactly the diagonal $\alpha = \beta$, where the two local overlaps coincide.

Table 3: Complete Uhlmann nodal set, Eqs. (23)–(25). Each Δ supports exactly Δ nodes; all satisfy $p = 1/2$ and $\alpha_k < \alpha_g$. The last column is the numerically evaluated $|Z_U|$.

(m, n)	Δ	k	C_k	μ_k	α_k	α_g	$ Z_U $
(0, 1)	1	0	0.5000	0.4472	0.5774	1.0000	4.5×10^{-16}
		1	0.7500	0.6000	0.9444	2.1094	1.8×10^{-16}
(0, 2)	2	0	0.2500	0.2425	1.6807	2.1094	6.1×10^{-17}
		1	0.7500	0.6000	0.9444	2.1094	1.8×10^{-16}
		2	0.2500	0.2425	1.6807	2.1094	6.1×10^{-17}
(0, 3)	3	0	0.1667	0.1644	2.8433	3.2756	3.4×10^{-15}
		1	0.5000	0.4472	2.1094	3.2756	1.8×10^{-16}
		2	0.8333	0.6402	1.3628	3.2756	2.1×10^{-15}
(1, 2)	1	0	0.5000	0.4472	0.4095	0.7221	5.7×10^{-15}
(1, 3)	2	0	0.2500	0.2425	1.3763	1.8000	6.1×10^{-17}
		1	0.7500	0.6000	0.7265	1.8000	6.6×10^{-14}
(1, 4)	3	0	0.1667	0.1644	2.5872	3.0354	5.5×10^{-15}
		1	0.5000	0.4472	1.8378	3.0354	5.0×10^{-15}
		2	0.8333	0.6402	1.1224	3.0354	2.4×10^{-15}
(2, 4)	2	0	0.2500	0.2425	1.1867	1.5829	3.1×10^{-15}
		1	0.7500	0.6000	0.6130	1.5829	1.8×10^{-16}

Eq. (13). The interference between branches is present in the connection but invisible in ϕ ; it takes a separately controllable branch phase to expose it.

Level 2 — entanglement enters affinely. In the balanced symmetric family the global Berry phase is an affine function of the concurrence, Eq. (14), interpolating between a mean-photon average and a pure interference term. This is as far as a pure-state geometric phase can go: it cannot be singular.

Level 3 — mixed-state holonomy, and it has two sides. Tracing out either mode leaves a genuinely mixed state, and the Uhlmann holonomy can be singular. The compact result (19) — parallel transport multiplies the off-diagonal generator by the concurrence — makes the structure transparent: at $p = 1/2$ the transported generator has magnitude ΔC , and the singularity condition $|b| = k + 1/2$ quantizes C to the finite ladder (23).

The most instructive feature is the interplay between what entanglement fixes and what it does not. The concurrence is shared by the two subsystems and determines *which* values are geometrically singular — the ladder C_k is a property of the pair. But it does not determine *where* those singularities sit, nor even whether the two sides have the same phase: the local populations p_A, p_B , equivalently the tilts θ_A, θ_B of the reduced Bloch vectors relative to each mode's own logical axis, are set by the local nonorthogonality g_α and h_β , which Schmidt's theorem leaves free. Symmetry is recovered exactly on the diagonal $\beta = \alpha$; and in the crossed family, where the local geometries do coincide but the indices are exchanged, one gets not symmetry but the exact antisymmetry $\Phi_U^A + \Phi_U^B = 0$. Entanglement is symmetric; local geometry need not be, and mixed-state holonomy sees both.

Finally, the critical amplitude α_g is not a thermodynamic critical point but an overlap-controlled boundary: it is fixed entirely by $|g_{mn}(\alpha_g)|^2 = 1/2$ and separates the region where equal logical-basis population is kinematically accessible from the region where it is not. Larger α makes the two photon-added branches more distinguishable, pushing the achievable population imbalance away from the symmetric point that an exact node requires.

11 Conclusions

Rebuilding the analysis of entangled PACS around the global two-mode cycle $U(\phi) = e^{-i(\hat{n}_A + \hat{n}_B)\phi}$ yields:

- (i) a closed-form global Berry phase equal to the total mean photon number, Eq. (12), exactly additive across modes and affine in the concurrence in the balanced symmetric family;
- (ii) the Uhlmann connection from the anticommutator transport condition, giving $(G + \Lambda)_{\text{od}} = C G_{\text{od}}$ and $b = \omega \sqrt{1 + C^2 - 4p(1 - p)}$, validated against two connection-free holonomy algorithms;
- (iii) a finite nodal ladder,

$$p = \frac{1}{2}, \quad C_k = \frac{2k + 1}{2\Delta}, \quad |g_{mn}(\alpha_k)|^2 = \frac{1 - C_k}{2}, \quad k = 0, \dots, \Delta - 1, \quad (30)$$

i.e. exactly Δ isolated Uhlmann singularities per family, common to both subsystems and all bounded by α_g ;

- (iv) a precise account of the $A \leftrightarrow B$ asymmetry: shared concurrence, unshared local populations, with symmetry on $\beta = \alpha$ and exact antisymmetry $\Phi_U^B = -\Phi_U^A$ in the crossed family.

Taken together these results provide an analytic bridge between photon addition, nonorthogonality, bipartite entanglement and mixed-state quantum holonomy, in a model where every step can be written — and independently checked — in closed form.

A Scalar products of photon-added coherent states

Inserting Eq. (3) into $\langle \alpha, m | \beta, n \rangle$ and using Fock orthogonality, only terms with a common level survive. Writing $\ell = n + p$ and $\Delta = n - m \geq 0$,

$$\langle \alpha, m | \beta, n \rangle = A_m A_n \alpha^\Delta \sum_{p=0}^{\infty} (\alpha\beta)^p \frac{(n+p)!}{(\Delta+p)! p!}, \quad A_n = \frac{e^{-\alpha^2/2}}{\sqrt{n! L_n(-\alpha^2)}}. \quad (31)$$

The key step is to recognize the factorial ratio as a ratio of Pochhammer symbols, $\frac{(n+p)!}{(\Delta+p)!} = \frac{n!}{\Delta!} \frac{(n+1)_p}{(\Delta+1)_p}$, which converts the sum into a confluent hypergeometric series and gives

$$\langle \alpha, m | \beta, n \rangle = \frac{e^{-\frac{1}{2}(\alpha^2 + \beta^2)} \alpha^\Delta n!}{\Delta! \sqrt{m! n! L_m(-\alpha^2) L_n(-\beta^2)}} {}_1F_1(n+1; \Delta+1; \alpha\beta) \quad (32)$$

for $0 \leq m \leq n$, the amplitude carrying the exponent Δ being the one attached to the *lower* index. Note that the ratio $n!/\Delta!$ arises whole from the Pochhammer reduction and therefore sits *outside* the square root. We verify Eq. (32) against direct Fock summation to 2×10^{-16} .

B Cross matrix element of the number operator

Repeating the sum with each term weighted by its Fock eigenvalue $\ell = n + p$,

$$\kappa_\alpha^{mn} = \langle \alpha, m | a^\dagger a | \alpha, n \rangle = A_m A_n \alpha^\Delta [n S(x) + x S'(x)], \quad x = \alpha^2, \quad (33)$$

where $S(x) = (n!/\Delta!) {}_1F_1(n+1; \Delta+1; x)$ is the sum of Appendix A. Using $\frac{d}{dx} {}_1F_1(a; b; x) = \frac{a}{b} {}_1F_1(a+1; b+1; x)$ together with a Kummer contiguous relation, this reduces to

$$\kappa_\alpha^{mn} = g_{mn}(\alpha) Q_{mn}(x), \quad Q_{mn}(x) = (n+1) \frac{{}_1F_1(n+2; \Delta+1; x)}{{}_1F_1(n+1; \Delta+1; x)} - 1. \quad (34)$$

Being a dimensionless ratio, Q_{mn} is insensitive to the prefactor conventions of Appendix A. Verified against Fock summation to 6×10^{-15} .

C Numerical conventions

All results use real amplitudes. Balanced plots take $\mu = \nu = 1/\sqrt{2}$; two-dimensional weight maps impose $\nu = \sqrt{1 - \mu^2}$ with $0 \leq \mu \leq 1$. Uhlmann plots show the principal value $\text{Arg } Z_U \in (-\pi, \pi]$. The shaded region above α_g in Figs. 6 and 8 is analytic, not a numerical cutoff: it is where $p = 1/2$ has no solution. Root-finding uses bisection on the monotonic $|g_{mn}(\alpha)|^2$; Laguerre polynomials and ${}_1F_1(a; b; z)$ are evaluated with `scipy.special`. The truncated Fock space uses 90 levels per mode, ample for $\alpha \lesssim 3$. The complete code accompanies this manuscript as `pacs_global_cycle.ipynb`.

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